OLEG TINKOV



AI / ML Platform Engineer | Python · LLM & RAG · Fintech
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Portfolio | GitHub | LinkedIn | ORCID

PROFESSIONAL SUMMARY

AI / ML Platform Engineer (Ph.D., Associate Professor) with 4+ years of experience combining backend software engineering, applied machine learning and LLM integrations. Currently an AI Engineer at Agroprombank, building on-premise RAG, OCR, speech-analytics and code-review solutions that run entirely inside the bank's perimeter. Hands-on background in credit scoring, signature verification and model interpretability (SHAP). Having trained and evaluated models myself, I design platform tooling and pipelines that data scientists actually adopt — with clean engineering practices, code reviews and reproducible deployments.

TECHNICAL SKILLS

Backend & Platform: Python (async, OOP), FastAPI, structured HTTP/REST APIs, error handling & retries, SQL, Docker, Linux, Git, code review, Streamlit

LLM & Applied AI: RAG architecture, embeddings, semantic chunking, context-window management, function/tool calling, AI agents, on-premise AI, OCR, speech analytics

Machine Learning: PyTorch, TensorFlow/Keras, scikit-learn, CatBoost / gradient boosting, SHAP, Pandas; classification, regression, scoring

Fintech & Domain: Credit-risk scoring, document (passport) validation, signature verification, call-center analytics, privacy-first deployment inside the bank perimeter

PROFESSIONAL EXPERIENCE

AI Engineer — Agroprombank, Tiraspol 11.2025 – Present

- Built a RAG-based assistant over the corporate knowledge base: semantic chunking, embeddings, context-window management and function/tool calling to connect internal knowledge sources with intelligent agents.
- Implemented an OCR system for client passport validation, supporting document checks in banking processes.
- Developed speech analytics for call-center conversations.
- Created an AI code-review agent to improve review quality and team productivity.
- Delivered solutions as asynchronous Python (FastAPI) services with structured HTTP APIs, failure/retry handling and SQL storage, containerized with Docker on Linux.
- Designed all solutions to run fully within the bank's internal perimeter with no external API calls, keeping customer data in-house.

Computational Chemist / Cheminformatician — Ligand Pro, Moscow 06.2025 – 12.2025

- Developed ML models for predicting drug activity and toxicity, from data preparation to evaluation.
- Built KRASAVA, a web expert system for virtual screening of KRAS G12D inhibitors; published in Int. J. Mol. Sci. (2025).

Associate Professor, Pharmacology — Shevchenko Transnistria State University 2020 – 09.2025

- Developed, validated and interpreted ML models (CatBoost, scikit-learn, Keras; SHAP-based and structural interpretation) for physiologically active substances and ADMET properties.
- Designed and deployed 8 public ML web services (Streamlit) that turn trained models into tools for non-programmers.
- Authored 37 journal articles, 40 conference proceedings and 1 book chapter; taught lectures and practical classes.

SELECTED PROJECTS

- **Credit AI Assistant** — credit-risk scoring with a Gradient Boosting model (AUC-ROC 0.7826), served as a web application. credit-ai-assistant.streamlit.app
- **DeepSign** — signature verification system (authentic vs. forged) built with TensorFlow and computer vision. deepsign.streamlit.app

- **ToxAI_assistant** — published ML platform combining database lookup and QSAR-model prediction of acute toxicity. tox-ai-assistant.streamlit.app
- **KRASAVA, HDAC1/2/3/6 services** — suite of ML web services for virtual screening of enzyme inhibitors; see portfolio.

EDUCATION & TRAINING

Ph.D. (Candidate of Chemical Sciences), Organic Chemistry — IPAC, Russian Academy of Sciences 2017
Master's degree, Chemistry — Shevchenko Transnistria State University 2001 – 2006
Courses: Deep Learning in Finance; SQL Introduction; KARPOV.COURSES SQL Simulator; Python Programming & OOP in Python; Introduction to Data Science and Machine Learning; Basics of Statistics (Stepik).

RESEARCH & PEER REVIEW

- **Selected publications:** KRASAVA — An Expert System for Virtual Screening of KRAS G12D Inhibitors, *Int. J. Mol. Sci.* 2025; ToxAI_assistant, *SAR QSAR Environ. Res.* 2025; HDAC3_VS_assistant, *Mol. Divers.* 2024.
- Peer reviewer since 2020 for 18 international journals, including *Molecular Cancer, Scientific Reports, Journal of Cheminformatics, BMC Bioinformatics, Toxicology Letters.*

Profiles: ORCID | ResearchGate | Web of Science ResearcherID: Q-4699-2017

LANGUAGES

Russian — native; English — intermediate (B1.3).